

# Unified Substrate Theory

Dustin Lee

The universe is not running hundreds of separate laws. It is running one mechanism. Every physical quantity: energy, momentum, force, charge, wavelength, flow, state change, curvature, and motion are patterns in a continuous medium or substrate. The geometry is a description of how much stuff there is, how fast it shears, and how strongly it curves.

- Matter is a density pattern.
- Motion is a shear pattern.
- Gravity is a curvature pattern.
- Charge is a displacement pattern.
- Waves are repeating patterns.
- State change is pattern update.
- All of physics is pattern  $\times$  flow  $\times$  pull.

Every constant we use — *c, h, G, e, particle masses* — drops directly into that structure with no exception.

## The Core Geometry

Matter moves according to three quantities:

- Mass-geometry — how much substrate is present
- Velocity-geometry — how fast the substrate is shearing
- Gravity-geometry — how strongly curvature is pulling the substrate

Every physical formula is just one way of sampling those three geometric fields.

The unified update is:

$$\Delta \text{State} = \text{flow} \times \text{pattern} \times \Delta S$$

This is the substrate's step-forward rule.

It is the same structure behind energy, waves, gravity, charge, wavelength, and motion.

## How the Standard Constants Fit the Structure

### Energy:

$$E = mc^2$$

Mass  $\times$  speed-limit<sup>2</sup>  $\rightarrow$  density pushed through maximum shear.

### Wave energy:

$$E = hf$$

Planck constant  $\times$  pattern frequency  $\rightarrow$  repeating shear.

### Gravity:

$$F = Gm_1m_2 / r^2$$

Curvature  $\times$  density  $\times$  density  $\rightarrow$  pull geometry.

### Charge work:

$$W = eV$$

Displacement  $\times$  pull  $\rightarrow$  directed motion.

### Wavelength:

$$\lambda = h / mc$$

Pattern spacing = Planck geometry / (density  $\times$  shear).

### State update:

$$\text{State\_new} = \text{State\_now} + (\text{flow} \times \text{pattern} \times \Delta S)$$

Every one of these is the same structure expressed through different constants.

## The Unified Statement

All physical behavior reduces to one sentence:

**Matter moves through a pattern under a pull.**

- Mass sets the density.
- Velocity sets the shear.
- Gravity sets the curvature.
- Charge sets the displacement.
- Planck sets the step size.
- Light speed sets the shear limit.
- The gravitational constant sets the curvature strength.

**Every formula in physics is just one way of measuring how much matter moves, how fast it moves, and how strongly it is pulled. That is unified physics. That is UST. That is the entire structure.**